

INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

THE SITE

Location QTS Manassas DC5, VA -- station KHEF, America/New_York
Committed pair OSM 1287260559 -> 1181391922
OSM way 1287260559 / Amazon IAD131
Facade gap 131.1 m between the two halls
Weather record 42,554 real hourly records from KHEF
Plume physics 576 steady-state solves on this site's own footprints; worst intake rise 0.4115 C at 165 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day 2023-04-11 (crossing)
Plant limit 18.0 C
Notice required 3 h
Forecast skill 0.50 relative to persistence
Level anchor sensor
Condenser bank longest facade
Switch budget 1 mode changes per day
Minimum dwell 3 h
Max dew point 15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 3 of 24 hours with 1 mode change. The reactive incumbent took 12 hours -- 9 more -- but declared 1 unsafe hour against the agent's 0. The incumbent had to break its own switch budget 1 time to stay safe; the agent never did. 11 hours were safe but still ran chillers, because the schedule could not afford them.

Agent 3 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent 12 of 24 hours free cooling, 2 mode change(s), 1 unsafe hour(s)
the incumbent broke its own switch budget 1 time(s) to stay safe
11 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	yes	switch budget	5.377	18.0	5.043	1.135
01	mechanical	yes	switch budget	5.273	18.0	4.104	1.343
02	mechanical	yes	switch budget	4.443	18.0	2.433	1.240
03	mechanical	yes	switch budget	4.723	18.0	2.994	1.198
04	mechanical	yes	switch budget	3.333	18.0	1.323	1.144

05	mechanical	yes	switch budget	1.968	18.0	0.213	1.111
06	mechanical	yes	switch budget	2.468	18.0	-0.397	1.416
07	mechanical	yes	switch budget	4.998	18.0	4.104	1.722
08	mechanical	yes	switch budget	8.888	18.0	9.104	2.451
09	mechanical	yes	switch budget	11.932	18.0	12.868	2.859
10	mechanical	yes	switch budget	16.388	18.0	16.884	3.007
11	mechanical	no	dry-bulb	19.317	18.0	20.109	2.299
12	mechanical	no	dry-bulb	21.668	18.0	22.994	1.837
13	mechanical	no	dry-bulb	23.474	18.0	24.244	1.503
14	mechanical	no	dry-bulb	24.904	18.0	24.839	1.389
15	mechanical	no	dry-bulb	26.217	18.0	25.560	1.284
16	mechanical	no	dry-bulb	25.516	18.0	25.000	1.260
17	mechanical	no	dry-bulb	25.069	18.0	24.446	1.473
18	mechanical	no	dry-bulb	24.444	18.0	22.805	1.741
19	mechanical	no	dry-bulb	21.938	18.0	18.544	1.953
20	mechanical	no	dry-bulb	20.278	18.0	15.774	2.066
21	FREE	yes	-	17.778	18.0	12.434	1.915
22	FREE	yes	-	15.273	18.0	11.323	1.796
23	FREE	yes	-	12.498	18.0	8.543	1.449

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 5.377 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

01:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 5.273 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

02:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.443 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

03:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.723 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

04:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 3.333 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a

thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

05:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 1.968 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 2.468 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.998 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

08:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 8.888 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

09:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 11.932 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

10:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.388 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.317 C against a 18.0 C limit, so it fails by 1.317 C. It would take a limit 1.317 C higher, or a bound 1.317 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.668 C against a 18.0 C limit, so it fails by 3.668 C. It would take a limit 3.668 C higher, or a bound 3.668 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.474 C against a 18.0 C limit, so it fails by 5.474 C. It would take a limit 5.474 C higher, or a bound 5.474 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.904 C against a 18.0 C limit, so it fails by 6.904 C. It would take a limit 6.904 C higher, or a bound 6.904 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 26.217 C against a 18.0 C limit, so it fails by 8.217 C. It would take a limit 8.217 C higher, or a bound 8.217 C tighter, to change this hour.

16:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.516 C against a 18.0 C limit, so it fails by 7.516 C. It would take a limit 7.516 C higher, or a bound 7.516 C tighter, to change this hour.

17:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.069 C against a 18.0 C limit, so it fails by 7.069 C. It would take a limit 7.069 C higher, or a bound 7.069 C tighter, to change this hour.

18:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.443 C against a 18.0 C limit, so it fails by 6.443 C. It would take a limit 6.443 C higher, or a bound 6.443 C tighter, to change this hour.

19:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.938 C against a 18.0 C limit, so it fails by 3.938 C. It would take a limit 3.938 C higher, or a bound 3.938 C tighter, to change this hour.

20:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.278 C against a 18.0 C limit, so it fails by 2.278 C. It would take a limit 2.278 C higher, or a bound 2.278 C tighter, to change this hour.

21:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.778 C, which is 0.222 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.915 C, and the margin is measured, not chosen: 1.575 C of group-conditional forecast error for this hour of day, plus 0.3399 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.273 C, which is 2.727 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.796 C, and the margin is measured, not chosen: 1.456 C of group-conditional forecast error for this hour of day, plus 0.3399 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 12.498 C, which is 5.502 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.449 C, and the margin is measured, not chosen: 1.109 C of group-conditional forecast error for this hour of day, plus 0.3399 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

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